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Simulation-informed transfer learning improves low-data nanobody thermal stability prediction

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Experimental measurements of nanobody melting temperature (T_m) are scarce, which limits machine-learning models of thermal stability. Simulations can provide more training labels, but these labels describe properties related to T_m rather than T_m itself. We asked how the choice of simulated variants and measured quantity affects their usefulness. We trained multi-task models that used a shared ESM2 sequence model for experimental T_m and one computed label at a time. We used 57 sequences for training, 114 for validation, and 396 for held-out testing. Each label and encoder setting was tuned separately using only the T_m validation set, then compared on the same test proteins. The choice of simulated variants mattered. An MD native-contact label reduced frozen-encoder T_m error by 0.20°C when computed for local mutation scans on two fixed structures, compared with 0.05°C for a mixed set of sequences. The measured quantity also mattered. Free-energy perturbation (FEP) labels improved both frozen and fine-tuned encoders, reducing the fine-tuned mean absolute error from 6.55°C to 6.40°C . Native-contact labels helped only the frozen encoder, while Rosetta and ThermoMPNN scores gave little or no improvement. These results show that simulation data for experimental prediction should be built from carefully chosen variants and from quantities that are closely related to the experimental property.

Key words: nanobody thermostability, melting-temperature prediction, multi-task transfer learning, protein language model, molecular simulation labels

◀ Significance ▶

Simulations can provide more protein-stability labels than experiments, but those labels do not always improve prediction of an experimental property. For nanobody T_m prediction, two choices mattered: which variants were simulated and which quantity was calculated. Native-contact labels helped more when computed for local mutation scans than for a mixed set of sequences. Free-energy labels helped both frozen and fine-tuned sequence models, whereas native-contact labels helped only the frozen model. More simulation is therefore not enough; the variants and the calculated quantity must be chosen carefully.

Introduction

Machine-learning models for molecular design need measured examples, but many experiments are slow or expensive. One way to increase the training data is to calculate related properties with quantum chemistry, molecular simulation, or

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other physical models. This approach is often called simulation-to-real (Sim2Real) learning. It has been used in materials science to combine large computed data sets with smaller experimental data sets [1–3]. In some cases, prediction improves as more computed data are added [3]. This outcome is not guaranteed. A calculation may use an imperfect model, sample the wrong systems, or measure a property that is only weakly related to the experiment.

Protein stability makes this problem clear because several common labels have different physical meanings. Melting temperature (T_m) is the experimental temperature at which folded and unfolded populations are balanced under a given assay. A mutation free-energy change ($\Delta\Delta G$) instead describes the change in folding free energy caused by a sequence change. Rosetta and learned stability models provide scores on their own scales, while molecular dynamics (MD) can provide structural measurements such as the fraction of native contacts that remain during a trajectory. All of these labels concern protein stability, but none is a T_m measurement. Adding more of the wrong label may therefore leave T_m prediction unchanged or make it worse.

We study this question in nanobodies, single-domain VHH antibodies used as small and modular binding proteins [4,5]. Thermal stability affects their expression, storage, and formulation [6–8]. Direct measurements and detailed simulations of stability remain costly [9–12]. Protein language models such as ESM2 can learn useful sequence features from large protein databases [13–15], and several methods use these features to predict T_m [16–21]. However, a sequence model does not remove the need for measured T_m values from the protein family of interest.

Large protein-stability data sets are becoming available. For example, Tsuboyama et al. measured folding stability for about 776,000 natural and designed protein domains by cDNA-display proteolysis [22]. Those domains are 40–72 residues long and differ from VHH domains in both sequence and measurement type. Models trained on small domains also transfer less well to longer proteins [23]. Nanobody-specific data and validation are therefore still needed [24–26].

The open question is not simply whether computed labels can be added to a T_m model. It is which variants should be simulated and which quantity should be calculated. A mixed set of unrelated sequences may contain unwanted differences such as sequence length. A mutation scan on one or two fixed structures gives a more controlled set of sequence changes. Even with the same variants, a mutation free energy, a Rosetta score, and an MD native-contact value may differ in how well they relate to T_m . These choices are made before model training, so they should be tested as part of the simulation plan.

Here, we compare several computed labels in the same low-data T_m task. The labels are mutation free energies from free-energy perturbation (FEP), Rosetta mutation scores, ThermoMPNN scores, Rosetta scores for ESM2-proposed or random variants, and MD native-contact values. For the MD label, we compare a mixed sequence set with local mutation scans on the same 1MEL and 4IDL structures used for FEP. Each computed label is trained as a second task that shares an ESM2 sequence model with T_m prediction (Fig. 1). We tune every label separately using only the experimental T_m validation set. We then compare the selected models on the same held-out T_m test proteins using paired errors. In this paper, “transfer” means that adding a computed label lowers the held-out T_m error.

Two results stand out. First, the choice of simulated variants matters. The MD native-contact label gives a much larger improvement with a frozen encoder when it is calculated for local mutation scans on fixed structures than for a mixed set of sequences. Second, the calculated quantity matters. FEP labels improve T_m prediction with both frozen and fine-tuned encoders, whereas native-contact labels help only the frozen encoder. Rosetta and ThermoMPNN scores give little or no improvement after separate tuning. These results show that useful simulation data require two choices before calculation: a controlled set of sequence changes and a quantity that is closely related to the experimental property.

Materials and methods

Study design

We asked whether computed protein-stability labels could improve nanobody melting-temperature (T_m) prediction when only a small experimental training set was available. Experimental T_m was always the task used to choose a model. Each computed label was added as a second regression task that shared all or part of the sequence model with the T_m task. We compared each label with a T_m -only model trained and selected in the same encoder setting.

All choices were made with the experimental T_m validation set. The test set was used only after the model form and training settings had been fixed. Errors on the computed labels were not used for checkpoint or hyperparameter selection.

Experimental T_m data

We used the thermo-seq task from NbBench, a public nanobody benchmark [25]. We reassigned its three published partitions before the comparisons in this study so that the smallest partition was used for training. The resulting split contained 57 training, 114 validation, and 396 test sequences. This split models the setting that motivated the study: few

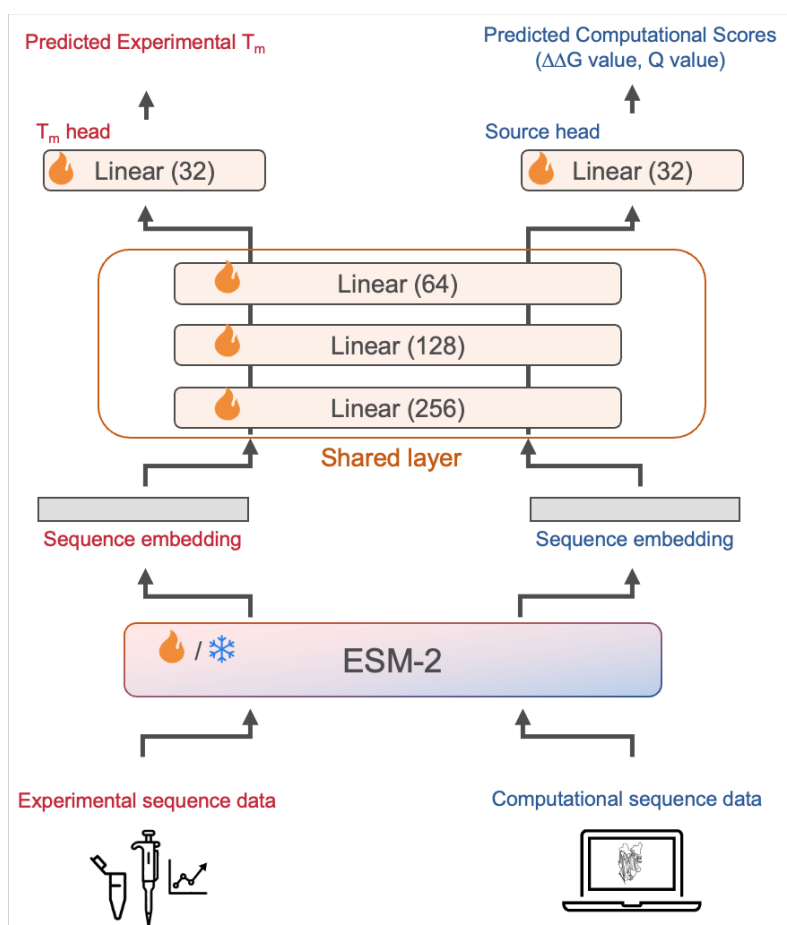


Figure 1 Multi-task model used to test computed stability labels. Experimental sequences and sequences with computed labels pass through the same ESM2 model and shared multilayer perceptron. Separate scalar heads predict T_m and the computed label. The ESM2 model is either kept frozen or fine-tuned. Models are selected using experimental T_m validation data, and the final comparison uses held-out T_m test data.

T_m measurements for fitting a model, but enough held-out proteins for paired comparisons. The training set fit model parameters, the validation set selected checkpoints and hyperparameters, and the test set was kept for final evaluation.

T_m values were stored in degrees Celsius. A pipeline consisting of a robust scaler followed by min–max scaling to [0,1] was fit on the 57 training values only. It was then applied to the training and validation labels. Predictions were returned to degrees Celsius before validation and test errors were calculated. Validation and test T_m values therefore did not affect the scaling step.

Computed-label tables and sampling

We tested seven sources of computed labels: FEP mutation free energies, matched MD native-contact values, Rosetta mutation scores, ThermoMPNN predictions, Rosetta scores for random variants, Rosetta scores for ESM2-proposed variants, and MD native-contact values for a diverse nanobody panel. These values were treated as related measurements, not as estimates of T_m.

FEP, Rosetta, and ThermoMPNN used two single-mutation tables based on the 1MEL and 4IDL structures. Each contained 435 and 409 rows, respectively. The matched MD tables used the same two structures and mutation scans and contained 431 1MEL rows and 406 4IDL rows that reached the common analysis time. The random and ESM2-proposed variant sets each contained 1000 two-mutation sequences per structure. The diverse MD table contained 1143 single-domain antibody sequences.

For all two-table sources, the notation n means a cap of n rows from *each* structure table, not n rows in total. Thus,

$n = 320$ gives at most 640 computed-label rows. Rows were sampled independently from the 1MEL and 4IDL tables with the run seed. The diverse MD source had one table, so n is the total number sampled from that table. After sampling, each computed-label table was divided at random into 80% training and 20% monitoring rows. These monitoring rows were not passed to the trainer for checkpoint selection; only the experimental Tm validation rows were used for that purpose.

The 1MEL and 4IDL tables were kept as two tasks because their score ranges can differ. Models used either two separate output heads or one output head with a learned structure indicator, as described below. Every processed table supplied an amino-acid sequence and one scalar label scaled to [0,1]. FEP, Rosetta, and ThermoMPNN scores were sign-reversed so that larger values indicated a more favorable change, then processed separately for each structure with robust scaling, a Yeo–Johnson transform, and min–max scaling. MD contact values were min–max scaled with larger values indicating greater contact retention.

We also checked exact sequence matches between every computed-label table and the NbBench splits. There were no exact matches for the FEP, matched MD, Rosetta, ThermoMPNN, random-variant, or ESM2-proposed tables. The diverse MD table contained 2 training, 4 validation, and 8 test sequences also present in NbBench. We retained these rows and report the overlap because it limits the interpretation of that comparison.

FEP mutation free energies

The FEP labels were mutation free-energy changes, $\Delta\Delta G$, for the 1MEL and 4IDL mutation scans. Positive raw $\Delta\Delta G$ denotes a destabilizing mutation. The learning tables therefore used the sign-reversed value before the scaling described above.

The fixed FEP tables were produced with alchemical simulations in NAMD using the CHARMM36 protein force field and explicit TIP3P water [27–29]. The starting coordinates were the 1MEL and 4IDL crystal structures. Mutation systems were prepared with the VMD AlaScan workflow and CHARMM hybrid-residue topologies [30, 31]. Representative archived input files show a 2-fs timestep, particle mesh Ewald electrostatics, bonds to hydrogen held fixed, a temperature of 300 K, and a pressure of 1 atm [32, 33].

Each forward and backward transformation covered 20 windows of width $\Delta\lambda = 0.05$ between 0 and 1. A window contained 500,000 steps (1 ns), with its first 100,000 steps assigned to equilibration. The folded-state free energy was combined with a residue-specific unfolded-state reference to obtain $\Delta\Delta G$ relative to the wild type. The present learning pipeline starts from the processed CSV files and does not rerun the FEP calculations.

MD native-contact labels

Both MD labels were based on the smooth native-contact fraction Q [34]. For a native contact with reference distance d_{ij}^0 and distance $d_{ij}(t)$ at time t , its contribution was

$$q_{ij}(t) = \left[1 + \exp \left\{ \beta [d_{ij}(t) - \lambda d_{ij}^0] \right\} \right]^{-1},$$

with $\beta = 50 \text{ nm}^{-1}$ and $\lambda = 1.8$. Native contacts were pairs within 0.45 nm in the reference frame and separated by more than three residues. Q was the mean over the selected contacts and frames.

Matched mutation scans. We simulated the 1MEL and 4IDL wild types and the same single-mutant sequences used in the FEP comparison. Each protein was placed in a cubic box of OPC water with a 15-Å buffer and 0.15 M salt, and was described by Amber ff19SB with hydrogen-mass repartitioning [35, 36]. OpenMM was used for minimization, equilibration, and production [37]. The recorded run settings were 0.2 ns NVT followed by 0.8 ns NPT at 300 K, 0.5 ns restrained NVT at 400 K, and restraint-free NVT production at 400 K. All stages used a 4-fs timestep; production coordinates were written every 10 ps. Although some runs continued to 100 ns, the label used the first 4000 production frames (40 ns) for every variant.

For this source, the reference distances came from production frame 0 and all backbone atoms selected by MDAnalysis were eligible for contacts. We computed $\Delta Q = Q_{\text{mutant}} - Q_{\text{WT}}$ within each structure and then min–max scaled it. Subtracting one constant wild-type value does not change a min–max-scaled label, so scaled Q and scaled ΔQ are identical within each structure table.

Diverse nanobody panel. The comparison panel was built from SAbDab structures [38]. One verified heavy-chain nanobody was taken per PDB entry when an IMGT-renumbered structure was available. Preparation used PDBFixer and pH-aware protonation with pdb2pqr/PROPKA when available [39, 40]. The systems used ff19SB, OPC water, 0.15 M salt, and OpenMM. After 300 K NVT and NPT equilibration, two restrained 400 K NVT stages preceded 100 ns of unrestrained 400 K NVT production. Requiring a usable topology and trajectory, a sequence of at least 50 residues, and at least one

selected contact gave 1143 rows.

The diverse-panel label used production frame 0 as its reference and averaged approximately the last 30 ns. Its contact set contained backbone heavy-atom pairs for which at least one residue was Asp, Glu, Gln, Asn, Arg, Lys, or His. It was min–max scaled across the single table. The matched and diverse studies used similar simulation time per sequence and therefore compare two ways of spending a similar computational budget. They used the same Best–Hummer form and the same temperature, although their contact selection and averaging window were not identical. We therefore treat this as a comparison of complete simulation plans, rather than as a one-variable test of sequence choice alone.

Rosetta, ThermoMPNN, and variant-selection comparisons

Rosetta scores were calculated with the `ddg_monomer` application and the `ref2015` score function [41,42]. The archived command used 50 iterations, Cartesian minimization, minimum-score aggregation, and full-atom input. The same settings were used for the 1MEL/4IDL mutation tables and for the two sets of two-mutation variants.

For the random set, mutation positions and replacement amino acids were sampled uniformly, excluding cysteine as a replacement, until 1000 unique two-mutation variants had been obtained for each structure. For the ESM2-guided set, the 650M-parameter ESM2 model generated 100,000 two-mutation variants per structure [14]. Variants were ranked by pseudo-log-likelihood, and the top 1% were scored with Rosetta. ThermoMPNN predictions were made for the same 435 and 409 single-mutation sequences used by the structure-matched tables [43]. The current loader reads the two processed ThermoMPNN tables, rather than the larger raw Colab outputs stored with them.

Sequence model and regression heads

The main encoder was the 8M-parameter ESM2 model `facebook/esm2_t6_8M_UR50D` [14]. Sequences were padded or truncated to 160 residues. The first-token representation was used as the sequence vector. In the shared model, this vector passed through linear layers of size 256, 128, and 32, with ReLU activation and dropout after the first two layers. The Tm head and each computed-label head were separate 32-to-1 linear layers.

The model search also included residual and latent forms. Both gave Tm and computed-label examples separate 256–128–32 paths. The residual form added a sigmoid-gated correction to a base Tm prediction. The latent form predicted a Tm correction from the Tm vector, the second path’s vector, and their elementwise product. For two-table sources, we compared separate 1MEL and 4IDL heads with a single head conditioned on a learned structure embedding. In the residual and latent implementations, the computed-label loss received a detached encoder vector by default, whereas the Tm path could still update the encoder. In the shared form, losses from all tasks could update the encoder when it was not frozen.

We tested two encoder settings. In the frozen setting, all ESM2 parameters were fixed and sequence vectors were computed once before head training. In the hot setting, all ESM2 parameters could change. AdamW used a smaller learning rate for ESM2 than for the newly initialized layers. Selected controls used 35M- and 650M-parameter ESM2 encoders.

Loss and model selection

Each row had one task identifier and contributed loss only through that task’s output head. We used Huber loss with $\delta = 1$ on the scaled labels. By default, the losses were combined with learned uncertainty weights [44]. For task k , loss L_k , and learned log-scale s_k , the term added to the total loss was

$$\frac{L_k}{2 \exp(2s_k)} + s_k.$$

Fixed task weights were tested as controls.

Models were trained with AdamW, cosine learning-rate decay, a batch size of 16, 100 warmup steps, and at most 400 epochs. Mixed-precision training was used on CUDA devices. A checkpoint was saved and evaluated once per epoch. The checkpoint with the lowest mean squared error on the 114 experimental Tm validation rows was retained, and training stopped after 15 epochs without improvement.

Every label source was tuned separately in the frozen and hot settings. The search had two stages. Stage 1 crossed three model forms (shared, residual, and latent) with two head forms (separate and structure-conditioned); Tm-only models searched the three model forms without a computed-label head. Stage 2 held the chosen form fixed. For hot models it tested encoder learning rates of 1×10^{-4} , 5×10^{-5} , 3×10^{-5} , and 1×10^{-5} . For frozen models it tested head learning rates of 1×10^{-3} , 5×10^{-4} , and 2×10^{-4} . Both searches crossed dropout values of 0.05, 0.15, and 0.30 with weight decay values of 0.02 and 0.10.

Each candidate was ranked by the MAE of a three-seed ensemble on the experimental Tm validation set, using $n = 320$

rows per structure table. The selected setting was then trained with five seeds for the final test result. Label-count curves used the same selected setting at $n = 20, 80, 160$, and 320 per structure; they were not retuned at each point.

Test evaluation and statistics

For each condition, the five models predicted all 396 test sequences. Their predictions were averaged in scaled space and then returned to degrees Celsius. The main measure was mean absolute error (MAE). We also calculated root mean squared error, R^2 , and Spearman and Pearson correlations.

Uncertainty was estimated by nonparametric bootstrap over test proteins. For one condition, we resampled its 396 absolute errors with replacement 10,000 times, using random seed 42, and calculated MAE for each sample. The reported 90% confidence interval is the 5th to 95th percentile range.

For a comparison, the same resampled test indices were applied to the candidate and reference error vectors. We calculated ΔMAE as candidate MAE minus reference MAE for every sample. Negative values therefore favor the candidate. Main figures report the mean paired ΔMAE and its 90% confidence interval. We do not use a separate p-value threshold for the main claims.

Software and saved outputs

The data loader, training code, model settings, and ensemble evaluation are in `prepare.py`, `train.py`, and the scripts distributed with the repository. Each final run stores its command-line settings, predictions, absolute errors, summary measures, and bootstrap results in machine-readable files. The figure scripts read those saved files directly. The released workflow begins with the fixed processed CSV tables; it does not repeat the raw FEP, Rosetta, ThermoMPNN, or MD calculations.

Results and discussion

Computed labels were judged only by experimental Tm

We first set up a common test for all computed labels (Fig. 1). Experimental Tm was the prediction target. FEP, MD, Rosetta, and ThermoMPNN values were separate regression tasks; they were not treated as extra Tm measurements. Each task used the same ESM2 sequence model and either shared or task-specific regression layers. We tested a frozen ESM2 encoder and a fully fine-tuned, or hot, encoder.

Only experimental Tm was used to choose a model. The 57 training proteins fit the model, the 114 validation proteins selected the architecture and training settings, and the 396 test proteins were used after selection. Every computed label was tuned separately in the frozen and hot settings. We then compared absolute errors on the same test proteins. This was necessary because a model can predict a computed value well without improving Tm.

The selected Tm-only model had a test MAE of 7.23°C with a frozen encoder and 6.55°C with a hot encoder. The lower error of the hot baseline shows that fine-tuning ESM2 was useful even with only 57 measured training proteins. Against these two baselines, we asked two questions. First, does the set of sequences chosen for simulation affect transfer? Second, does the computed quantity affect whether the benefit remains when ESM2 is fine-tuned?

Local mutation scans gave a stronger MD result

Figure 2a compares two choices made before running MD. The heterogeneous screen used unrelated single-domain antibody sequences of different lengths. The matched scan used local mutations of the fixed 1MEL and 4IDL structures. Both calculations used 400 K MD and the Best-Hummer native- contact fraction Q . They also used similar simulation time per sequence. Thus, the comparison asks how a similar amount of simulation should be distributed over sequence space.

The heterogeneous and matched studies were run and tuned as separate series, so their absolute MAEs cannot be compared directly. Figure 2b instead shows the paired change from the Tm-only model in each series. In the heterogeneous series, the MD label changed MAE by -0.049°C with a frozen encoder (90% CI, -0.085 to -0.012°C) and by $+0.117^\circ\text{C}$ with a hot encoder (90% CI, $+0.012$ to $+0.224^\circ\text{C}$). The frozen gain was small, and the hot model became worse. In the matched series, the frozen change was -0.195°C (90% CI, -0.336 to -0.055°C), about four times the change in the heterogeneous series. The hot change was $+0.029^\circ\text{C}$, with a confidence interval that included zero.

Several differences between the two sequence sets may contribute to this result. The heterogeneous set ranged from 58 to 461 residues, and its Q value had a Pearson correlation of $r = +0.13$ with sequence length. Length could therefore provide an unwanted shortcut. The matched scans kept length fixed within each structure and supplied local directions: one mutation and its change in contact retention. This is a simpler learning problem. However, the present data do not

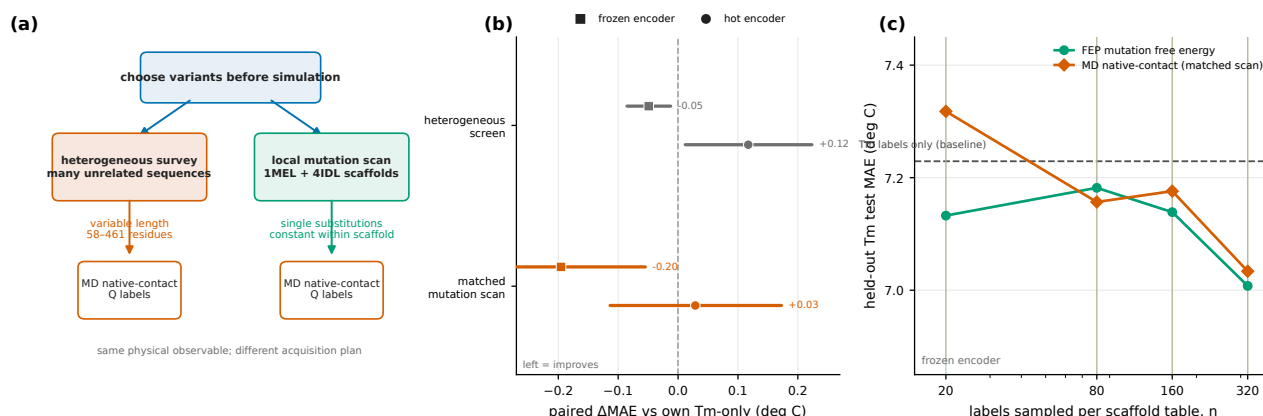


Figure 2 Simulation design changes the value of an MD native-contact label. (a) A heterogeneous screen of variable-length sequences and local mutation scans on the fixed 1MEL and 4IDL structures use similar simulation time per sequence and produce labels from the Best-Hummer Q family. (b) Paired Δ MAE for each design and encoder setting. Because the two series were tuned independently, each value is relative to that series' own T_m-only model; their absolute MAEs are not compared. The matched scan gives a larger frozen benefit and avoids the hot penalty, but gives no hot benefit. (c) Frozen-encoder MAE as the number of FEP or matched-MD labels increases. The dashed line is the selected frozen T_m-only model. Here n is the number sampled from each structure table. Points and bars show ensemble MAE and 90% bootstrap confidence intervals.

show that length alone caused the difference. The two MD pipelines also differed slightly in contact selection and in the trajectory frames used for averaging, as detailed in Materials and Methods. The result therefore compares the two complete simulation plans, not a single changed variable.

Wild-type referencing does not explain the matched result. Its label was $\Delta Q = Q_{\text{mutant}} - Q_{\text{WT}}$, but the same wild-type constant was subtracted from every mutation in a structure table. The following min-max scaling removes this constant, so scaled Q and scaled ΔQ are identical within that table.

The frozen label-count curves give a second view of the matched scan (Fig. 2c). Here n is the number of labels sampled from each structure table. FEP MAE changed from 7.13°C at $n = 20$ to 7.01°C at $n = 320$. Matched MD changed from 7.32 to 7.03°C and fell below the 7.23°C T_m-only baseline at the largest setting. The four points are not monotonic enough to support a scaling law, but both sources were better than T_m-only at $n = 320$. Their paired changes were -0.22°C for FEP and -0.20°C for MD. The direct FEP-minus-MD difference was only -0.03°C (90% CI, -0.21 to +0.16°C). Thus, with a frozen encoder, the matched MD scan performed as well as FEP at the largest tested label count.

The main conclusion from Fig. 2 is narrow. Choosing local mutations on fixed structures made the frozen MD result much stronger and removed the hot penalty seen for the heterogeneous screen. It did not make the MD label helpful in the hot model. Sequence choice matters, but it is not the only choice that matters.

Only FEP helped both frozen and hot models

We next compared all computed labels after separate tuning (Fig. 3a). With a frozen encoder, FEP and matched MD gave the clearest gains. FEP reached 7.01°C MAE, a paired change of -0.221°C from T_m-only (90% CI, -0.366 to -0.077°C). Matched MD reached 7.03°C, a change of -0.195°C. ThermoMPNN was third at 7.09°C, but its change of -0.141°C had a confidence interval that included zero. Plain Rosetta, random variants scored by Rosetta, and ESM2-proposed variants scored by Rosetta were close to or worse than the T_m-only model.

The order changed with a hot encoder. FEP reached 6.40°C, the lowest MAE in the study. Its paired change from the 6.55°C T_m-only model was -0.153°C (90% CI, -0.295 to -0.009°C). No other computed label improved the hot baseline. Matched MD, ThermoMPNN, plain Rosetta, and random variants plus Rosetta had changes from +0.029 to +0.144°C, all with intervals that included zero. ESM2-proposed variants plus Rosetta were worse by +0.411°C (90% CI, +0.181 to +0.651°C).

The focused comparisons in Fig. 3b separate the two all-atom labels. FEP and matched MD were not distinguishable with a frozen encoder, as noted above. With a hot encoder, FEP was better than MD by a paired -0.182°C (90% CI, -0.359 to -0.009°C). This difference also appears across label counts (Fig. 3c). At $n = 20$, matched MD was worse than

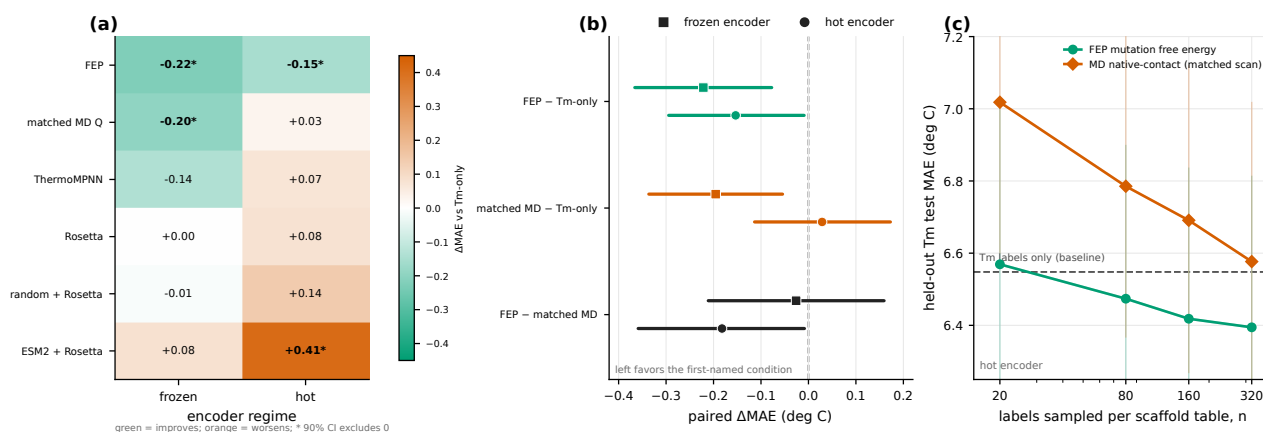


Figure 3 FEP is the only computed label that improves both encoder settings. (a) Paired Δ MAE from the selected Tm-only model for every tuned label. Green denotes lower error and orange higher error. An asterisk marks a 90% paired bootstrap confidence interval that excludes zero. (b) Paired comparisons of FEP and matched MD with Tm-only, and direct FEP-minus-MD comparisons, in the frozen and hot settings. Values left of zero favor the first-named condition. (c) Hot-encoder MAE as FEP or matched-MD labels increase. FEP becomes better than Tm-only, whereas matched MD recovers from its low-count loss only to the baseline. Here n is the number sampled from each structure table. Bars are 90% bootstrap confidence intervals over the 396 test proteins.

hot Tm-only by $+0.47^{\circ}\text{C}$ (90% CI, $+0.29$ to $+0.65^{\circ}\text{C}$). Its error then fell from 7.02 to 6.58°C as n increased, but at $n = 320$ it had only returned to the Tm-only baseline. FEP fell from 6.57 to 6.40°C and was better than baseline at $n = 320$. These four-point curves show recovery and crossover; they do not establish a power law.

FEP $\Delta\Delta G$ and Tm are different measurements. $\Delta\Delta G$ gives the change in folding free energy caused by a mutation, whereas Tm is the temperature at which folded and unfolded states are balanced under an experimental condition. They are nevertheless linked by protein folding thermodynamics. In this data set, the measured Tm examples supply absolute temperature values across nanobodies, while the FEP scans supply relative information about which local sequence changes stabilize or destabilize two structures. The matched MD scan also supplies local information, but contact retention is a more indirect measure of folding stability. This difference is a plausible reason why the two labels tie in the frozen model but separate in the hot model.

We describe this pattern as a difference in transfer depth. Here, “deeper” means only that a benefit remains when the pretrained encoder is allowed to change. A recent preprint from our group gives a more concrete basis for this interpretation [45]. That study used sparse autoencoders to examine ESM2 after fine-tuning on nanobody Tm. It found interpretable features linked to surface charge, hydrophobic-core packing, the VHH tetrad, and disulfide bonds. Thus, fine-tuning on nanobody Tm can produce a representation with specific, physically meaningful stability features rather than only a better regression fit.

Together with that result, the present frozen-hot difference suggests a testable hypothesis: mutation free energies may support the same kinds of local stability features that a Tm-fine-tuned encoder learns, while a single contact-retention value may provide a weaker guide for updating them. This remains an interpretation, not a result measured here. We did not apply sparse autoencoders or compare hidden states in the present models, and the selected FEP and MD models did not always use the same regression architecture. A direct next test is therefore to compare the sparse features learned by Tm-only, Tm+FEP, and Tm+MD hot encoders and ask which stability features appear, disappear, or change strength.

Controls, scope, and next tests

Several controls limit alternative explanations. Increasing ESM2 from 8M to 35M or 650M parameters did not reproduce the 8M FEP result (Supplementary Fig. 4), so a larger pretrained model was not a substitute for the computed label. An analytical correction for periodic-image effects in charge-changing FEP mutations had almost no effect on the scaled labels. Tests that clipped the largest FEP values or added the available Phe scan did not improve transfer. These checks support the use of the original FEP tables, although they do not prove that every large $\Delta\Delta G$ value is quantitatively accurate.

The other label sources provide an important negative result. ThermoMPNN gave at most a weak frozen gain, and none of the Rosetta sources improved both encoder settings. ESM2-proposed variants scored by Rosetta did not beat random

variants scored by the same method. We therefore treat these sets as comparisons of sequence choice and score type. The results do not support a claim that the ESM2 proposals form a successful protein-design loop.

The findings agree with the main idea of Sim2Real learning: computed data are useful when they share information with the measured task [3]. The protein case adds an earlier decision. Before a label can transfer, the simulated sequences must make the desired relationship easy to learn. A large mixed set is not automatically better than a smaller local scan. After the sequences are chosen, the computed quantity still matters. FEP and matched MD used roughly similar simulation time per variant here, but we did not record wall-clock cost in a form that supports a general cost claim.

This study has several limits. The Tm training set contains only 57 nanobodies, and the matched labels cover only two structures. The heterogeneous and matched MD pipelines differ in more than their sequence sets, and the heterogeneous table includes eight exact matches to the test set. The source list does not cover every useful simulation measurement. The confidence intervals also show that small differences among the middle-ranked labels remain uncertain. Finally, we did not make new Tm measurements or test predictions prospectively.

The next test should extend matched mutation scans to more nanobody structures while keeping the simulation and label-extraction settings fixed. It should compare several physical quantities on exactly the same variants and then test the selected Tm model on new experimental measurements. Such a study would separate sequence choice from label choice more cleanly and show whether the FEP result extends beyond 1MEL and 4IDL.

Conclusion

Computed labels improved low-data nanobody Tm prediction only in specific settings. The sequences chosen before simulation mattered: MD native-contact labels from local mutation scans on 1MEL and 4IDL gave a clearer frozen-encoder gain than labels from a heterogeneous sequence screen. The calculated quantity also mattered. FEP mutation free energies improved both frozen and fine-tuned models, whereas matched MD native-contact values improved only the frozen model; Rosetta and ThermoMPNN scores gave no clear gain. Simulation data for experimental prediction should therefore be planned around controlled sequence changes and a quantity closely related to the measured property, rather than around data volume alone. The next test is to extend matched scans to more structures and evaluate the resulting models on new Tm measurements.

In the longer term, the same idea could guide an iterative design study. A model could choose which variant to calculate next, receive an FEP or MD result, and use that result to choose the following variant. Reinforcement learning is one way to make these sequential choices, but a computed score alone would not be a reliable reward. Our ESM2-proposed variants did not outperform random variants when both were scored by Rosetta, showing that a generator and a physics-based score do not by themselves make a useful design method. Any future reinforcement-learning study should therefore be compared with random sampling and simpler active-learning methods, and its success should be judged prospectively by new experimental Tm measurements.

Conflict of interest

T.M. is an employee of Epsilon Molecular Engineering, Inc. Y.M. is a scientific advisor of Epsilon Molecular Engineering, Inc.

Author contributions

Y.M. conceived and designed the study. Y.M. and T.M. carried out the overall computational study and analyzed the data. S.O. and K.O. performed the FEP calculations. Y.M. and T.M. wrote the manuscript. All authors discussed the results, reviewed the manuscript, and approved the final version.

Data availability

The source code for data processing, model training, analysis, and figure generation is available at <https://github.com/ymatsunaga/sim2real-nanobody-tm>. The processed data used in this study, together with reduced MD trajectories sufficient to reproduce the reported labels, will be available from Zenodo (DOI: 10.5281/zenodo.XXXXXXX). Full raw simulation files are not included because of their size and are available from the corresponding author upon reasonable request.

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